

Rational Herglotz Companion Values

Dyadic descent, complete formal classification,
and explicit dilogarithm identities

Research continuation prepared for the ProveIt project

OpenAI-assisted mathematical research note

October 2026

Abstract

For the Herglotz companion $J(x) = \int_0^1 \log(1+t^x)/(1+t) dt$, we resolve the manuscript's all-rational *formal* reduction problem. At a reduced mixed-parity ratio, with even entry e and odd entry o , reduction to logarithms is equivalent to $e^2 \equiv \pm 1 \pmod{o}$ and $o^2 \equiv 1 \pmod{2e}$. Allowing rational dilogarithms adds precisely six rational arguments: $1/3, 3, 1/5, 5, 3/5, 5/3$. The proof combines a conductor-doubling identity with rigidity of the new dyadic character layer. It uses the existing all-conductor symbol theorem, whose proof is recalled.

We give a complete even-coefficient quadratic-recurrence parametrization of the logarithmic cases and an all-finite-order height-counting expansion. Analytically, a Hecke relation and the Rogers five-term identity yield an explicit log-sine formula for every $J(n/(n+1))$, and a compact evaluation of $J(3/5)$. A prime-seven exterior-logarithm witness disproves a finite-field three-term assertion in the reviewed Radchenko–Zagier preprint, without affecting the direct boundary formula used here. Exact finite tests and independent high-precision integral comparisons accompany the proofs. Formal non-reducibility is not numerical transcendence; no period-injectivity or proof-assistant claim is made.

Source snapshot. `cc34f73596336f2466d9754cb0f3635bd2bedade`

Target. `Analysis/Polylogarithms/docs/manuscript`

Package. LaTeX source, compiled PDF, exact arithmetic checks, independent numerical diagnostics, theorem ledger and integration notes.

Contents

1	The question being continued	2
1.1	Results and their dependencies	2
1.2	Scope of the audit and claims of novelty	2
2	The exact symbol calculus and the established kernel theorem	3
2.1	Rationalized exterior symbols	3
2.2	Cyclotomic symbols and the direct boundary formula	3
2.3	A faithful model on the symbol span	4
2.4	Separating coprime conductors	5
3	Conductor doubling and rigidity of the new dyadic layer	6
3.1	Odd-conductor doubling	6
3.2	Even-conductor descent is rigid	6
3.3	Two finite-group difference tests	7
4	Complete formal classification at rational arguments	7
4.1	Examples and the missing factor of two	9
5	Even quadratic recurrences and height asymptotics	9
5.1	A complete hierarchy of height terms	10
6	Explicit analytic identities with full constants	11
6.1	Classical normalized functional equations	12
6.2	Every neighboring rational argument	12
6.3	The exceptional value at $3/5$	14
6.4	All six non-logarithmic formal exceptions are explicit	15
7	An exact correction to a finite-field assertion	15
7.1	Precise correction and unaffected statements	16
7.2	Editorial changes recommended for the manuscript	16
8	Reproducibility, exact tests and numerical diagnostics	17
8.1	Exact finite checks	17
8.2	Independent defining-integral comparisons	17
8.3	Files, environment and replay	18
9	Further research questions	18
9.1	Short complete certificates for every recurrence pair	18
9.2	Other prime filters	19
9.3	Integral structure and torsion	19
9.4	Sharper exact counting and uniform depth	19
9.5	Period evaluation versus formal obstruction	19
9.6	A corrected cocycle formulation and colored extensions	19
9.7	Formal verification targets	20
	Conclusion	20
	References	21

1 The question being continued

The current ProveIt manuscript treats special polylogarithmic values, cyclotomic relations, exact certificates and Herglotz arithmetic in one framework. Its Herglotz chapter proves a complete formal classification of $F(p/q) - F(1)$ and of $J(2/q)$. Its discovery chapter explicitly leaves the general $J(p/q)$ symbol problem as a further direction [1]. That is the problem addressed here. The Gaussian reductions, the S_4 certificate and the other completed continuations are not reclaimed as new work.

For $x > 0$, use the standard normalization

$$F(x) = \sum_{n \geq 1} \frac{\psi(nx) - \log(nx)}{n}, \quad J(x) = \int_0^1 \frac{\log(1+t^x)}{1+t} dt. \quad (1.1)$$

Radchenko–Zagier establish the finite cyclotomic dilogarithm formula for rational F and the identity

$$J(x) = F(2x) - 2F(x) + F(x/2) + \frac{\pi^2}{12x}. \quad (1.2)$$

Their modified Herglotz function has a Hecke relation that will supply explicit evaluations in Section 6 [2]. The broader two-parameter Herglotz–Zagier–Novikov function and its specializations are studied by Choie–Kumar [3].

There are two different questions. An analytic evaluation is an equality of numbers with specified branches. A formal reduction asks whether the finite dilogarithm expression admits reduction by the rationalized five-term calculus. Section 2 defines the latter exactly. The classification below is complete for that calculus. It does not classify every conceivable numerical equality among these periods.

1.1 Results and their dependencies

The new algebraic step is not another numerical rank computation. At an odd conductor, doubling the root set gives an exact three-term *conductor* identity for exterior symbols. At an even conductor, a symbol in the new dyadic layer cannot be absorbed by symbols from the preceding field unless it is zero. These facts reduce the boundary of $J(p/q)$ to two sparse finite-group calculations.

The resulting classification has a particularly simple shape. For mixed parity, logarithmic and rational-dilogarithmic reducibility coincide. For odd numerator and denominator, the only nontrivial rational-dilogarithmic cases are the ratios between 1, 3 and 5. The condition modulo $2e$, not merely modulo e , is essential.

A Vieta descent then parametrizes all logarithmic ratios using the recurrences $w_{j+1} = cw_j \pm w_{j-1}$ with c even. This gives both an exact enumerator and a sparse height law. Finally, a separate analytic argument yields a uniform formula for neighboring ratios and evaluates the exceptional value $J(3/5)$. These analytic identities carry their full logarithmic and π^2 terms; they are not merely boundary certificates.

1.2 Scope of the audit and claims of novelty

The reference snapshot is `cc34f73596336f2466d9754cb0f3635bd2bedade`. The main source file is `chapters/09-herglotz.tex`, whose Git blob is `b372a62797ae9161ddcfd9f4edd036ecfd87df36`. The dependency and open-question checks also included `chapters/10-discovery.tex` and the existing `herglotz-cyclotomic-obstructions` report.

Statements described as new are continuations relative to this reviewed snapshot, not claims of exhaustive bibliographic priority. Classical functional equations, the existing all-conductor symbol

theorem, and rationalized algebraic K -theory facts are attributed as inputs. Ordinary proofs are supplied for the new results. The accompanying finite tests are implementation checks, not substitutes for these proofs.

Section 7 gives one precise correction to a statement in the cited preprint. It does not purport to audit every assertion of that paper or of the complete ProveIt manuscript. In particular, it is not a claim that the manuscript's independently derived boundary formula is incorrect.

2 The exact symbol calculus and the established kernel theorem

2.1 Rationalized exterior symbols

For a number field K , put

$$\mathcal{A}(K) = K^\times \otimes_{\mathbb{Z}} \mathbb{Q}, \quad \langle uv \rangle = \langle u \rangle + \langle v \rangle.$$

All exterior powers in this article are over $\mathbb{Q}$. Roots of unity vanish in $\mathcal{A}(K)$. Embeddings of fields induce injective maps on these vector spaces and on their exterior powers, so we identify old-field symbols with their images in larger fields.

Definition 2.1 (Formal reductions). Let $\mathcal{P}(K)$ be the rationalized pre-Bloch group: the vector space on symbols $[z]$, modulo five-term relations, with $[0] = [1] = 0$. Write

$$\partial[z] = \langle z \rangle \wedge \langle 1 - z \rangle \quad (z \neq 0, 1), \quad \mathcal{B}(K) = \ker \partial.$$

For a Galois-invariant element $\eta \in \mathcal{P}(K)$, a *formal logarithmic reduction* means $\eta = 0$. A *formal rational-dilogarithmic reduction* means that η belongs to the image of $\mathcal{P}(\mathbb{Q})$.

The terminology suppresses products of logarithms of algebraic numbers and rational multiples of π^2 in evaluation. Compatible complex logarithms are allowed in this definition. The worked analytic identities below use only real logarithms of positive algebraic numbers and Li_2 on $(0, 1)$.

For finite Galois number fields, the standard rationalized inputs are

$$\mathcal{B}(K)^{\text{Gal}(K/\mathbb{Q})} = \mathcal{B}(\mathbb{Q}) = 0, \quad \partial\mathcal{P}(\mathbb{Q}) = \Lambda^2\mathcal{A}(\mathbb{Q}). \quad (2.1)$$

These are Bloch-group descent, the torsion of $\mathcal{B}(\mathbb{Q})$ and the torsion of $K_2(\mathbb{Q})$; the last assertion uses the identification of the Steinberg quotient with K_2 for a field. They are the same inputs used in [2, §7.2]. For the Bloch-group and K -theoretic framework, see [4, 5]. Consequently, for invariant η ,

$$\eta = 0 \iff \partial\eta = 0, \quad (2.2)$$

$$\eta \in \text{im } \mathcal{P}(\mathbb{Q}) \iff \partial\eta \in \Lambda^2\mathcal{A}(\mathbb{Q}). \quad (2.3)$$

For the reverse implication in the second line, choose a rational pre-Bloch element with the same boundary and apply the first line to the invariant difference. This makes clear where the established K -theory input enters; it is not replaced by a numerical observation.

2.2 Cyclotomic symbols and the direct boundary formula

For $m \geq 3$ and $(a, m) = 1$, define

$$\beta_m(a) = \sum_{k=1}^{m-1} \langle 1 - \zeta_m^{ak} \rangle \wedge \langle 1 - \zeta_m^k \rangle, \quad \zeta_m = e^{2\pi i/m}. \quad (2.4)$$

Use $\beta_1 = \beta_2 = 0$. The excluded term $k = 0$ is important: there is no multiplicative-group element $\langle 0 \rangle$. Reindexing gives

$$\beta_m(-a) = \beta_m(a), \quad \beta_m(a^{-1}) = -\beta_m(a). \quad (2.5)$$

Every $\beta_m(a)$ is Galois invariant. The inverse is taken modulo m .

For coprime positive p, q , the finite formula of [2, Theorem 3] defines an invariant element $\xi_{p/q} \in \mathcal{P}(\mathbb{Q}(\zeta_{pq}))$ representing the dilogarithmic part of $F(p/q) - F(1)$. Explicitly it is

$$\xi_{p/q} = \sum_{\alpha^p=1} \left(\sum_{\substack{\beta^q=1 \\ \beta \neq 1}} h(\alpha, \beta) - \sum_{\substack{\beta^p=1 \\ \beta \neq 1}} h(\alpha, \beta) \right), \quad h(\alpha, \beta) = \left[\frac{\beta}{\beta-1} \right] - \left[\frac{\alpha-\beta}{1-\beta} \right]. \quad (2.6)$$

Proposition 2.2 (Established boundary formula). *For coprime $p, q > 0$,*

$$\partial \xi_{p/q} = \beta_q(p) - \beta_p(q) - \langle p \rangle \wedge \langle q \rangle. \quad (2.7)$$

Proof. This is the boundary used in both [2] and [1]; we recall its direct verification to avoid relying on the assertion audited in Section 7. Terms with $\alpha = 1$ or $\alpha = \beta$ have zero boundary. For the others set $A = \langle 1 - \alpha \rangle$, $B = \langle 1 - \beta \rangle$, $C = \langle \alpha - \beta \rangle$. Since roots of unity vanish, direct expansion gives

$$\partial h(\alpha, \beta) = -C \wedge A + C \wedge B + B \wedge A.$$

The subtracted double sum in (2.6) has zero boundary by antisymmetry under $\alpha \leftrightarrow \beta$. In the first double sum, the product identities

$$\prod_{\substack{\beta^q=1 \\ \beta \neq 1}} (\alpha - \beta) = \frac{\alpha^q - 1}{\alpha - 1}, \quad \prod_{\substack{\alpha^p=1 \\ \alpha \neq 1}} (\beta - \alpha) = \frac{\beta^p - 1}{\beta - 1}$$

make the first two wedge contributions $-\beta_p(q)$ and $\beta_q(p)$. The last contribution is $\langle q \rangle \wedge \langle p \rangle$, since the products of $1 - \alpha$ and $1 - \beta$ over the nontrivial roots are p and q . Coprimality ensures that no additional common root was omitted. $\square$

2.3 A faithful model on the symbol span

Put $G_m = (\mathbb{Z}/m\mathbb{Z})^\times / \{\pm 1\}$ for $m \geq 3$. The following theorem is an existing result of the manuscript [1]; its proof is included because the new layer argument uses its Gram matrix, not just its conclusion.

Theorem 2.3 (Existing all-conductor kernel theorem). *For $c_a \in \mathbb{Q}$,*

$$\sum_{a \in G_m} c_a \beta_m(a) = 0 \quad \Longleftrightarrow \quad c_a = c_{a^{-1}} \quad (a \in G_m). \quad (2.8)$$

Thus the map on the symbol span

$$\beta_m(a) \mapsto \mathbf{e}_a - \mathbf{e}_{a^{-1}} \in \mathbb{Q}[G_m] \quad (2.9)$$

is faithful, and

$$\beta_m(a) = 0 \quad \Longleftrightarrow \quad a^2 \equiv \pm 1 \pmod{m}. \quad (2.10)$$

Proof. For $1 \leq k < m$ define vectors and a matrix

$$v_k = (\log |1 - \zeta_m^{bk}|)_{b \in G_m}, \quad A_m = \sum_{k=1}^{m-1} v_k v_k^\top.$$

Let $(P_a v)_b = v_{ba}$. Then $P_a v_k = v_{ak}$, so A_m commutes with all P_a .

The vectors v_k span the regular representation. Indeed their span is invariant, and it has a nonzero projection to each character. For a nontrivial even character of primitive conductor $f \mid m$, use $k = m/f$. Its Fourier coefficient is a positive integer multiple of

$$\sum_{b \in G_f} \chi^*(b) \log |1 - \zeta_f^b| = -\frac{1}{2} \tau(\chi^*) L(1, \overline{\chi^*}) \neq 0.$$

The equality follows by Abel-regularizing $\log |1 - re^{i\theta}| = -\sum_{n \geq 1} r^n \cos(n\theta)/n$ and applying the primitive Gauss-sum identity. Nonvanishing uses the nonzero primitive Gauss sum and Dirichlet's theorem $L(1, \chi) \neq 0$ for nonprincipal characters [6, 25.15.9]. The trivial character occurs using $k = m/\ell$ for a prime divisor ℓ of m : the logarithmic product is $\log \ell$; for $\ell = 2$ the vector is the nonzero constant vector $\log 2$. All nontrivial roots, not just primitive m th roots, are used. Therefore A_m is positive definite.

Apply the exterior square of the logarithm map $\mathcal{A}(\mathbb{Q}(\zeta_m)) \rightarrow \mathbb{R}^{G_m}$, and identify $v \wedge w$ with $vw^\top - wv^\top$. Its value on a symbol is

$$\ell_\wedge(\beta_m(a)) = P_a A_m - A_m P_a^\top = A_m (P_a - P_{a^{-1}}). \quad (2.11)$$

A zero combination of symbols therefore gives $\sum_a (c_a - c_{a^{-1}}) P_a = 0$ after multiplication by A_m^{-1} . The regular permutation matrices are linearly independent, so the coefficients are symmetric. Conversely symmetric coefficients cancel by (2.5); self-inverse classes contribute zero over $\mathbb{Q}$. This proves all assertions. $\square$

Remark 2.4. The exterior logarithm map is not asserted to be injective on all of $\Lambda^2 \mathcal{A}(K)$. Its explicitly computed value suffices on the particular span in Theorem 2.3. Positivity above is proved by character nonvanishing, not by a floating-point determinant.

2.4 Separating coprime conductors

Lemma 2.5 (Normalized norm separation). *Let $(r, s) = 1$, with U a rational combination of the $\beta_r(a)$ and V a rational combination of the $\beta_s(b)$, viewed in a common cyclotomic field. Then*

$$U - V \in \Lambda^2 \mathcal{A}(\mathbb{Q}) \iff U = V = 0.$$

Moreover, normalized norm to $\mathbb{Q}$ kills each such symbol combination.

Proof. The linear normalized field norm to E is $N_E \langle u \rangle = \langle \text{Norm}_{K/E} u \rangle / [K : E]$. It fixes old-field elements; apply its exterior square to wedges. Coprime cyclotomic fields are linearly disjoint over $\mathbb{Q}$. The norm of either factor of a term of $\beta_s(b)$ to $\mathbb{Q}(\zeta_r)$ is the same rational vector, since the factors are Galois conjugates in $\mathbb{Q}(\zeta_s)$. Thus that wedge is killed, while every old r -conductor wedge is fixed. Reversing the roles gives the other projection. If $U - V = W$ is rational, the two projections give $U = W$ and $-V = W$; projecting either again to $\mathbb{Q}$ gives $W = 0$. The cases of conductor 1 or 2 follow from the zero-symbol conventions. The same conjugate-factor argument proves the last assertion. $\square$

3 Conductor doubling and rigidity of the new dyadic layer

This section contains the new structural input. Doubling an odd conductor does not enlarge its cyclotomic field, whereas doubling an even conductor does. The two cases must be treated differently.

3.1 Odd-conductor doubling

Theorem 3.1 (Exact conductor-doubling identity). *Let m be odd and $(a, 2m) = 1$. In $\Lambda^2 \mathcal{A}(\mathbb{Q}(\zeta_m))$,*

$$\beta_{2m}(a) = 3\beta_m(a) - \beta_m(2a) - \beta_m(a/2), \quad (3.1)$$

where $a/2$ denotes multiplication by the inverse of 2 modulo m . The formula also holds for $m = 1$ under the zero-symbol convention.

Proof. For $m > 1$ split μ_{2m} into μ_m and $-\mu_m$. The first part of the sum defining $\beta_{2m}(a)$ is $\beta_m(a)$. Set $u_k = \langle 1 - \zeta_m^k \rangle$ for $k \not\equiv 0 \pmod{m}$. For these k ,

$$\langle 1 + \zeta_m^k \rangle = u_{2k} - u_k.$$

Since a is odd, the second part, apart from the zero wedge at the root -1 , is

$$\sum_{k=1}^{m-1} (u_{2ak} - u_{ak}) \wedge (u_{2k} - u_k).$$

The two diagonal terms each give $\beta_m(a)$ by reindexing. The cross terms give $-\beta_m(2a)$ and $-\beta_m(a/2)$. Including the first part proves (3.1). No logarithm branch is used in this multiplicative-group calculation. $\square$

3.2 Even-conductor descent is rigid

Theorem 3.2 (New dyadic layer rigidity). *Let m be even and $(a, 2m) = 1$. Then*

$$\beta_{2m}(a) \in \Lambda^2 \mathcal{A}(\mathbb{Q}(\zeta_m)) \iff \beta_{2m}(a) = 0 \iff a^2 \equiv \pm 1 \pmod{2m}. \quad (3.2)$$

Proof. For $m = 2$, all the symbols at conductor 4 vanish, so the assertion is immediate. Suppose $m \geq 4$. The natural map $G_{2m} \rightarrow G_m$ has kernel $\{1, h\}$, where h is the class of $1 + m$. This is a nontrivial element of order 2.

In the real regular representation $\mathbb{R}^{G_{2m}}$, logarithm vectors coming from $\mathbb{Q}(\zeta_m)$ lie in the $+1$ eigenspace of P_h . Consequently, a skew matrix arising from an exterior product of such vectors annihilates the -1 eigenspace. If $\beta_{2m}(a)$ descends, (2.11) therefore gives

$$A_{2m}(P_a - P_{a^{-1}})(I - P_h) = 0.$$

The Gram matrix is invertible. The faithful regular representation of the group algebra then gives

$$(\mathbf{e}_a - \mathbf{e}_{a^{-1}})(\mathbf{e}_1 - \mathbf{e}_h) = 0. \quad (3.3)$$

If $a \neq a^{-1}$ in G_{2m} , the coefficient of $\mathbf{e}_a$ in the left side is positive: the negative term $-\mathbf{e}_{a^{-1}}$ is distinct, $-\mathbf{e}_{ah}$ is distinct because $h \neq 1$, and any coincidence with $+\mathbf{e}_{a^{-1}h}$ only increases that coefficient. Hence (3.3) is impossible. Thus $a = a^{-1}$ in G_{2m} , which is exactly the last congruence in (3.2). Theorem 2.3 now makes the symbol zero. The converse implication to descent is immediate. $\square$

The theorem is stronger than merely testing the normalized norm of the new symbol. It detects a nonzero new-character component, so no combination of old-field wedges can cancel it. This is the obstruction that forces a modulus $2e$ in the final classification.

3.3 Two finite-group difference tests

For an odd conductor m , division by 2 in symbol arguments is well-defined. The following two tests will isolate the denominator and numerator contributions of J .

Lemma 3.3 (First difference). *For odd $m \geq 1$ and $(a, m) = 1$,*

$$\beta_m(a) - \beta_m(a/2) = 0 \iff m \in \{1, 3, 5\}. \quad (3.4)$$

Proof. The case $m = 1$ is immediate. For $m \geq 3$, put $x = \bar{a}$ and $b = \bar{2}$ in G_m , and let $f_x = \mathbf{e}_x - \mathbf{e}_{x^{-1}}$. By Theorem 2.3, the required equality is $f_x = f_{x/b}$. If this vector is nonzero, its unique positive basis coefficient identifies its index, so $x = x/b$ and $b = 1$. If both vectors are zero, $x^2 = (x/b)^2 = 1$, so $b^2 = 1$. In either case $b^2 = 1$, or $4 \equiv \pm 1 \pmod{m}$, forcing odd m to be 1, 3 or 5. Conversely the groups at 3 and 5 have exponent at most 2, and all their symbols vanish. $\square$

Lemma 3.4 (Second difference). *For odd $m \geq 1$ and $(a, m) = 1$,*

$$\beta_m(2a) - 2\beta_m(a) + \beta_m(a/2) = 0 \iff a^2 \equiv \pm 1 \pmod{m}. \quad (3.5)$$

Proof. Again $m = 1$ is automatic. In the group algebra of G_m , the image of the left side is $(T_b + T_{b^{-1}} - 2I)f_x$, where T_b is translation by $b = \bar{2}$. With the orthonormal basis $\mathbf{e}_g$,

$$\langle f, (2I - T_b - T_{b^{-1}})f \rangle = \|f - T_b f\|^2.$$

Vanishing of the second difference therefore forces $f_x = T_b f_x$. If $f_x \neq 0$, the unique positive coefficient of f_x forces $b = 1$. For an odd conductor this means $m = 1$ or 3, at which $f_x = 0$ already, a contradiction. Thus $f_x = 0$, and the kernel theorem gives the asserted congruence. Conversely $f_x = 0$ makes its second difference zero. The two positive and two negative translated terms agree with the symbol expression even though a single translation of f_x need not itself have the form f_y . $\square$

4 Complete formal classification at rational arguments

For positive rational x , reduce every subscript to lowest terms and set

$$\eta_x = \xi_{2x} - 2\xi_x + \xi_{x/2}. \quad (4.1)$$

By (1.2), this is the dilogarithmic element relevant to $J(x)$. The extra rational multiple of π^2 has no boundary.

Lemma 4.1 (Analytic and formal reciprocity). *For $x > 0$,*

$$J(x) + J(1/x) = \log^2 2. \quad (4.2)$$

For positive rational x , one also has $\eta_{1/x} = -\eta_x$ in a common pre-Bloch group. Hence each type of formal reduction is invariant under $x \mapsto 1/x$.

Proof. Integration by parts gives

$$J(x) = \log^2 2 - \int_0^1 \log(1+t) \frac{xt^{x-1}}{1+t^x} dt.$$

The substitution $u = t^x$ identifies the integral with $J(1/x)$. This known reciprocity also follows from the functional equations discussed in [3].

For the formal statement, (2.7) gives $\partial \xi_{1/r} = -\partial \xi_r$. Their invariant sum is zero by (2.2). Applying this to the three terms of (4.1) proves the assertion. Thus the formal assertion is not inferred solely from an equality of numerical periods. $\square$

Theorem 4.2 (Complete all-rational classification). *Let p, q be coprime positive integers.*

Mixed parity. *Let e denote the even entry and o the odd entry of (p, q) . Then $J(p/q)$ has a formal rational-dilogarithmic reduction if and only if it has a formal logarithmic reduction, and these are equivalent to*

$$\boxed{e^2 \equiv \pm 1 \pmod{o}, \quad o^2 \equiv 1 \pmod{2e}.} \quad (4.3)$$

Congruences modulo 1 are interpreted as automatic.

Both entries odd. *There is a formal rational-dilogarithmic reduction exactly when*

$$p, q \in \{1, 3, 5\}. \quad (4.4)$$

There is a formal logarithmic reduction exactly when $p = q = 1$.

Consequently, the rational-dilogarithmic but not logarithmic arguments are precisely

$$\boxed{\frac{1}{3}, 3, \frac{1}{5}, 5, \frac{3}{5}, \frac{5}{3}}. \quad (4.5)$$

Proof. For mixed parity, Lemma 4.1 lets us take $x = e/o$. Expanding (2.7) gives

$$\partial\eta_{e/o} = D_o(e) - N_e(o), \quad (4.6)$$

where

$$D_o(e) = \beta_o(2e) - 2\beta_o(e) + \beta_o(e/2), \quad (4.7)$$

$$N_e(o) = \beta_{2e}(o) - 2\beta_e(o) + \beta_{e/2}(o). \quad (4.8)$$

The rational wedges cancel because $-\langle 2e \rangle + 2\langle e \rangle - \langle e/2 \rangle = 0$. The conductors o and $2e$ are coprime. By Lemma 2.5, the boundary is rational if and only if both $D_o(e)$ and $N_e(o)$ vanish. It is then zero, proving that rational-dilogarithmic and logarithmic reducibility coincide.

Lemma 3.4 gives $D_o(e) = 0 \iff e^2 \equiv \pm 1 \pmod{o}$. If $N_e(o) = 0$, its top-conductor term belongs to $\Lambda^2 \mathcal{A}(\mathbb{Q}(\zeta_e))$. Theorem 3.2 forces $\beta_{2e}(o) = 0$, or $o^2 \equiv \pm 1 \pmod{2e}$. Since $4 \mid 2e$ and o is odd, the negative sign is impossible. Conversely $o^2 \equiv 1 \pmod{2e}$ kills all three symbols of $N_e(o)$, including the conductor-1 or conductor-2 degeneracies. This proves (4.3).

Now suppose both p and q are odd. Applying Theorem 3.1 at both conductors to the boundary expansion yields

$$\begin{aligned} \partial\eta_{p/q} &= (\beta_q(p) - \beta_q(p/2)) - (\beta_p(q) - \beta_p(q/2)) \\ &\quad + \langle 2 \rangle \wedge \langle p/q \rangle. \end{aligned} \quad (4.9)$$

For example, the rational contribution before simplification is $-\langle 2p \rangle \wedge \langle q \rangle + 2\langle p \rangle \wedge \langle q \rangle - \langle p \rangle \wedge \langle 2q \rangle$, which is the last term displayed. Normalized norm separation and Lemma 3.3 show that this boundary is rational exactly when $p, q \in \{1, 3, 5\}$.

For logarithmic reduction, normalized norm to $\mathbb{Q}$ first requires $\langle 2 \rangle \wedge \langle p/q \rangle = 0$. If the reduced odd ratio p/q is not 1, the prime-exponent vector of p/q is nonzero and has zero coordinate at the prime 2. It is therefore independent of $\langle 2 \rangle$ in $\mathcal{A}(\mathbb{Q})$, making this wedge nonzero. If $p = q = 1$, all terms vanish and $J(1) = \frac{1}{2} \log^2 2$. This proves the odd case and the list (4.5). $\square$

4.1 Examples and the missing factor of two

The mixed pairs $(e, o) = (4, 15), (12, 5), (12, 29)$ satisfy (4.3). Thus $J(4/15)$, $J(5/12)$ and $J(12/29)$ have formal logarithmic reductions. Explicit short logarithmic expressions for arbitrary pairs of this kind are a separate constructive task; the neighboring-ratio family will be evaluated below.

By contrast $(e, o) = (8, 3)$ satisfies $8^2 \equiv 1 \pmod{3}$ and $3^2 \equiv 1 \pmod{8}$, but fails $3^2 \equiv 1 \pmod{16}$. The new-layer obstruction can be seen exactly in G_{16} : with $h = 9$, the four-term expression (3.3) at $a = 3$ is

$$(\mathbf{e}_3 - \mathbf{e}_5)(\mathbf{e}_1 - \mathbf{e}_7) = 2\mathbf{e}_3 - 2\mathbf{e}_5 \neq 0,$$

where residues have been identified up to sign. Hence $J(8/3)$ has no formal rational-dilogarithmic reduction. Testing only modulo e would produce a false positive.

For $x = 2/q$ the theorem recovers exactly the existing classification. Odd q gives $q = 3, 5$; a denominator divisible by 4 reduces to a ratio with even entry and odd entry 1; and a denominator $q = 2m$ with odd m gives the odd ratio $1/m$, allowing precisely $m = 1, 3, 5$ for rational dilogarithms. Thus the positive cases are $q = 2, 3, 5, 6, 10$ or $4 \mid q$; the cases 6, 10 are not logarithmic in the formal calculus.

Remark 4.3 (What has not been proved). Theorem 4.2 is a necessary and sufficient condition for a five-term certificate in Definition 2.1. For a ratio that fails the test, it does not prove that the real number $J(p/q)$ is linearly independent of all logarithmic products or rational dilogarithm values. Such a claim would require information about numerical period evaluation beyond the symbol theorem. In particular, the numerical transcendence problems in the manuscript remain separate.

5 Even quadratic recurrences and height asymptotics

The modular criterion admits a complete elementary parametrization. It also gives a way to enumerate the exceptional rational set without factoring either entry.

Theorem 5.1 (Recurrence parametrization). *Apart from $J(1)$, all unordered reduced logarithmic pairs $1 \leq a < b$ are exactly the adjacent pairs in the following families, for every even integer $c \geq 2$:*

$$u_0 = 1, \quad u_1 = c, \quad u_{j+1} = cu_j - u_{j-1}, \tag{5.1}$$

$$v_0 = 1, \quad v_1 = c, \quad v_{j+1} = cv_j + v_{j-1}. \tag{5.2}$$

Both a/b and b/a are logarithmic cases. The seed $(1, c)$ is common to the two families. Beyond the seeds there is no overlap between the families or between different values of c .

Proof. Let e be even and o odd in a pair satisfying (4.3). First suppose $e^2 \equiv 1 \pmod{o}$. Then

$$c = \frac{e^2 + o^2 - 1}{eo}$$

is an integer. It is even: after division of the numerator by e , both e and $(o^2 - 1)/e$ are even, and the remaining divisor o is odd. Also $c \geq 2$, since $(e - o)^2 \geq 1$. Writing the sorted pair as $a < b$ gives

$$a^2 + b^2 - cab = 1. \tag{5.3}$$

If $a > 1$, the new entry $d = (a^2 - 1)/b = ca - b$ is a positive integer strictly smaller than a . The pair (d, a) has the same invariant, is coprime, and has opposite parity because c is even and d has

the parity of b . The invariant itself gives both congruences in (4.3). Repetition reaches a pair $(1, c)$. Reversing the descent is (5.1).

Next suppose $e^2 \equiv -1 \pmod{o}$. The signed integer

$$H = \frac{o^2 - e^2 - 1}{eo}$$

is even by the same divisibility argument, and is nonzero: otherwise $(o - e)(o + e) = 1$, impossible for $e > 0$. Put $c = |H| \geq 2$. In sorted coordinates,

$$b^2 - cab - a^2 = \varepsilon, \quad \varepsilon = \begin{cases} 1, & b \text{ odd,} \\ -1, & b \text{ even.} \end{cases} \quad (5.4)$$

For $a > 1$ set $d = b - ca = (a^2 + \varepsilon)/b$. This is a positive integer less than a ; for $\varepsilon = 1$, use $a^2 + 1 < a(a + 1) \leq ab$. The pair (d, a) is coprime and has opposite parity. Its invariant has sign $-\varepsilon$, exactly as required when the parity of the larger entry changes. Thus the descent continues to $(1, c)$, and its reverse is (5.2).

Conversely the recurrences preserve (5.3) or (5.4), respectively, and preserve coprimality and alternating parity. In the minus family the invariant gives $e^2 \equiv 1 \pmod{o}$ and $o^2 - 1 = e(co - e)$, which is divisible by $2e$. In the plus family it gives $e^2 \equiv -1 \pmod{o}$ and $o^2 - 1 = e(e \pm co)$, again divisible by $2e$. Thus every generated pair satisfies the criterion.

If the odd entry exceeds 1, the signs $+1$ and -1 modulo that entry cannot both hold. The displayed invariants determine c uniquely within each case, and the strictly decreasing descent identifies a unique depth. The only possible shared pairs are the seeds, whose odd entry is 1. $\square$

Family	Coefficient	Initial terms
Minus	2	1, 2, 3, 4, 5, 6, ...
Minus	4	1, 4, 15, 56, 209, ...
Minus	6	1, 6, 35, 204, 1189, ...
Plus	2	1, 2, 5, 12, 29, 70, ...
Plus	4	1, 4, 17, 72, 305, ...
Plus	6	1, 6, 37, 228, 1405, ...

The coefficient-2 minus family explains why every neighboring ratio is logarithmic. All other families grow geometrically. In particular, a descent algorithm can shortcut consecutive integers and otherwise uses $O(\log \max(a, b))$ steps. This statement counts arithmetic descent steps; it is not a bound on the bit complexity of an arbitrary expanded analytic certificate.

5.1 A complete hierarchy of height terms

Let $N(B)$ count unordered coprime pairs $1 \leq a < b \leq B$ satisfying the logarithmic criterion; $J(1)$ is not counted. Here B is a positive integer.

Theorem 5.2 (Height expansion to every fixed order). *For every fixed integer $R \geq 2$, as $B \rightarrow \infty$,*

$$N(B) = \frac{3}{2}B + \sum_{r=2}^R B^{1/r} + O_R(B^{1/(R+1)} \log B). \quad (5.5)$$

In particular,

$$N(B) = \frac{3}{2}B + \sqrt{B} + O(B^{1/3} \log B). \quad (5.6)$$

The number of ordered positive rational logarithmic arguments of height at most B is $2N(B) + 1$. For $B \geq 5$, the number admitting a formal rational-dilogarithmic reduction is $2N(B) + 7$.

Proof. Count each seed $(1, c)$ once: there are $\lfloor B/2 \rfloor$. The minus family at coefficient 2 contributes all consecutive pairs; after removing its seed it contributes $B - 2$ for $B \geq 2$. Their sum is $\frac{3}{2}B + O(1)$.

For the other pairs write $w_r^\pm(c)$ for the term at index r in the two recurrences. The pair with this height has indices $(r - 1, r)$. At fixed $r \geq 2$, induction on the recurrence gives

$$w_r^\pm(c) = c^r \pm (r - 1)c^{r-2} + O_r(c^{r-4}),$$

where the remainder can simply be read as the lower-degree polynomial terms; when r is small it is zero as appropriate. Equivalently, $w_r^\pm(c) = c^r + O_r(c^{r-2})$. The number of even coefficients satisfying $w_r^\pm(c) \leq B$ is therefore $\frac{1}{2}B^{1/r} + O_r(1)$: the transition can differ from $c = B^{1/r}$ by only $O_r(B^{-1/r})$, with finitely many small coefficients absorbed in the constant. Removing coefficient 2 from the minus family changes this by $O(1)$. Both signs together contribute $B^{1/r} + O_r(1)$. Summing over $2 \leq r \leq R$ gives the displayed fractional powers.

For completeness, the remaining depths are uniformly bounded. In the minus family with $c \geq 4$, induction on the successive ratios gives $w_j/w_{j-1} > c - 1$, so $w_j \geq (c - 1)^j$. In the plus family with $c \geq 2$, $w_j \geq c^j$. A term with index at least $R + 1$ and height at most B therefore requires $c \leq 1 + B^{1/(R+1)}$. There are $O(B^{1/(R+1)})$ possible coefficients, and each contributes only $O(\log B)$ depths, because its growth factor is at least 2. This proves the remainder estimate. Disjointness from Theorem 5.1 prevents overcounting.

Finally each unordered pair yields two reciprocal arguments, while $x = 1$ adds one. The six distinct extra arguments in (4.5) all have height at most 5. $\square$

The theorem implies a sparse set among all reduced rational arguments of bounded height, but no transcendence statement is attached to that sparsity. It counts exactly the formal reduction cases.

B	$N(B)$	Ordered logarithmic arguments
100	161	323
1 000	1 543	3 087
10 000	15 135	30 271
1 000 000	1 501 162	3 002 325
100 000 000	150 010 658	300 021 317
10 000 000 000	15 000 102 700	30 000 205 401

These are exact counts from the recurrence enumerator, not floating-point fits to (5.5). The script uses integer arithmetic and does not enumerate all $O(B^2)$ pairs at the larger heights.

6 Explicit analytic identities with full constants

The formal classification establishes existence of reductions but does not, by itself, print their logarithmic corrections. This section does so for an infinite family and for the remaining exceptional ratio $3/5$. All the identities in this section are ordinary analytic equalities. No integer-relation search is used in their proofs.

6.1 Classical normalized functional equations

For $0 < t < 1$ define the Rogers dilogarithm

$$\mathcal{L}(t) = \text{Li}_2(t) + \frac{1}{2} \log t \log(1-t), \quad \mathcal{L}(1/2) = \frac{\pi^2}{12}. \quad (6.1)$$

We use its reflection and five-term identities in the real interval:

$$\mathcal{L}(t) + \mathcal{L}(1-t) = \frac{\pi^2}{6}, \quad (6.2)$$

$$\mathcal{L}(A) + \mathcal{L}(B) = \mathcal{L}(AB) + \mathcal{L}\left(\frac{A(1-B)}{1-AB}\right) + \mathcal{L}\left(\frac{B(1-A)}{1-AB}\right), \quad 0 < A, B < 1. \quad (6.3)$$

These are classical; their normalization and relation to the ordinary dilogarithm are reviewed in [4, 2].

Put

$$f(x) = F(x) - F(1) + \frac{\pi^2}{12}(x - 2 + x^{-1}) - \frac{1}{4} \log^2 x. \quad (6.4)$$

The functional equations needed below are the normalized inversion and three-term equations and the determinant-2 Hecke equation of [2, (15), (22a)]:

$$f(1/x) = -f(x), \quad (6.5)$$

$$f(x) - f(x+1) - f\left(\frac{x}{x+1}\right) = \mathcal{L}(1/2) - \mathcal{L}\left(\frac{x}{x+1}\right), \quad (6.6)$$

$$f(2x) + f(x/2) + f\left(\frac{x+1}{2}\right) + f\left(\frac{2x}{x+1}\right) - 3f(x) = \mathcal{L}\left(\frac{x}{x+1}\right) - \mathcal{L}(1/2) + \frac{1}{2} \log 2 \log x. \quad (6.7)$$

They hold for $x > 0$ with real logarithms. Formula (1.2) becomes

$$J(x) = f(2x) - 2f(x) + f(x/2) + \frac{1}{2} \log^2 2 - \frac{\pi^2}{24}(x - x^{-1}). \quad (6.8)$$

Define the finite log-sine sum

$$S(m) = \sum_{k=1}^{m-1} \log^2 \left(2 \sin \frac{\pi k}{m} \right), \quad S(1) = 0. \quad (6.9)$$

Every sine argument in this sum is positive. The integer formula of [2, (23)] simplifies under (6.4) to

$$f(m) = \frac{(m-1)\pi^2}{24} + \frac{1}{2} S(m) + \frac{1}{4} \log^2 m, \quad m \geq 1. \quad (6.10)$$

These classical inputs are kept separate from the deductions that follow.

6.2 Every neighboring rational argument

Theorem 6.1 (Uniform neighboring-ratio identity). *For every integer $n \geq 1$,*

$$J\left(\frac{n}{n+1}\right) = \frac{1}{2} \log^2 2 + \frac{1}{2} (S(n) - S(n+1) + S(2n+2) - S(2n)) - \frac{\pi^2(n^2 - n - 1)}{24n(n+1)}.$$

(6.11)

The inverse ratios are evaluated by (4.2).

Proof. Solve (6.7) for the second difference in (6.8). Set

$$x = \frac{n}{n+1}, \quad A = \frac{2n+1}{2n+2}, \quad B = \frac{2n}{2n+1}, \quad C = \frac{n}{2n+1}.$$

The three f -terms on the resulting right side are $f(x) - f(A) - f(B)$. For any integer $m \geq 1$, (6.6) gives

$$f\left(\frac{m}{m+1}\right) = f(m) - f(m+1) - \mathcal{L}(1/2) + \mathcal{L}\left(\frac{m}{m+1}\right).$$

The integer terms telescope, leaving

$$\begin{aligned} f(2x) - 2f(x) + f(x/2) &= f(n) - f(n+1) + f(2n+2) - f(2n) \\ &\quad + \mathcal{L}(x) - \mathcal{L}(A) - \mathcal{L}(B) + \mathcal{L}(C) + \frac{1}{2} \log 2 \log x. \end{aligned} \quad (6.12)$$

In (6.3) with these A, B , the three right-hand arguments are respectively x , $1/2$ and C . Hence the Rogers combination in (6.12) is exactly $-\mathcal{L}(1/2)$.

Substitution of (6.10) contributes $\pi^2/24$ from the four integer terms. Their ordinary log-square contribution is

$$\frac{1}{4}(\log^2 n - \log^2(n+1) + \log^2(2n+2) - \log^2(2n)) = -\frac{1}{2} \log 2 \log x,$$

which cancels the cross term. The remaining π^2 coefficient in (6.8) is

$$\frac{1}{24} - \frac{1}{12} - \frac{1}{24}(x - x^{-1}) = -\frac{n^2 - n - 1}{24n(n+1)}.$$

This proves (6.11). All Rogers arguments used in the proof are strictly between zero and one, so no branch adjustment or unspecified constant is hidden in the cancellation. $\square$

Corollary 6.2 (Small explicit logarithmic evaluations). *Let $\phi = (1 + \sqrt{5})/2$. Then*

$$J(4/3) = \frac{7}{8} \log^2 2 - \frac{1}{2} \log^2(1 + \sqrt{2}) + \frac{5\pi^2}{288}, \quad (6.13)$$

$$J(4/5) = \frac{7}{8} \log^2 2 + 2 \log^2 \phi - \frac{1}{2} \log^2(1 + \sqrt{2}) - \frac{11\pi^2}{480}. \quad (6.14)$$

Proof. Write $a = \log 2$, $b = \log 3$, $c = \log 5$, $d = \log \phi$ and $u = \log(1 + \sqrt{2})$. Elementary trigonometric products give

$$\begin{aligned} S(3) &= \frac{1}{2}b^2, & S(4) &= \frac{3}{2}a^2, & S(5) &= \frac{1}{4}c^2 + d^2, \\ S(6) &= a^2 + \frac{1}{2}b^2, & S(8) &= \frac{7}{4}a^2 + u^2, & S(10) &= a^2 + \frac{1}{4}c^2 + 5d^2. \end{aligned}$$

For instance $2 \sin(\pi/10) = \phi^{-1}$ and $2 \sin(3\pi/10) = \phi$, while the two distinct primitive positive logarithms at conductor 8 have sum $a/2$ and difference u . Substitute into (6.11) for $n = 3, 4$, using reciprocity for $n = 3$. $\square$

At $n = 1, 2$, the theorem also recovers $J(1/2) = \pi^2/48 + \log^2 2/4$ and $J(2/3) = -\pi^2/144 + 3 \log^2 2/4$. These are consistency checks against known special cases, not additional novelty claims.

6.3 The exceptional value at $3/5$

Theorem 6.3 (A compact exceptional dilogarithm identity). *With $\phi = (1 + \sqrt{5})/2$,*

$$\boxed{J(3/5) = \text{Li}_2(1/3) - \frac{1}{2} \text{Li}_2(1/5) + \frac{1}{2} \log^2 2 + \frac{1}{2} \log^2 3 - \frac{1}{4} \log^2 5 + \log^2 \phi - \frac{7\pi^2}{180}.} \quad (6.15)$$

In particular $[1/3] - \frac{1}{2}[1/5]$ is a rational-dilogarithm core.

Proof. We first track the normalized Herglotz terms, then remove the remaining Rogers terms by a single five-term substitution. Set $a = \log 2$, $b = \log 3$, $c = \log 5$, $d = \log \phi$. The classical $F(2/5)$ identity [2, (24)] is

$$F(2/5) - F(1) = \frac{1}{2} \text{Li}_2(4/5) - \frac{1}{5} \pi^2 + \log^2(2 \sin(\pi/5)) + \log^2(2 \sin(2\pi/5)) + a \log(2/5).$$

The two log-sine squares sum to $c^2/8 + d^2/2$. Converting to f and to the Rogers normalization gives

$$f(2/5) = \frac{1}{2} \mathcal{L}(4/5) - \frac{1}{8} \pi^2 + \frac{3}{4} a^2 - \frac{3}{8} c^2 + \frac{1}{2} d^2. \quad (6.16)$$

The integer formula gives

$$\begin{aligned} f(2) &= \pi^2/24 + 3a^2/4, & f(3) &= \pi^2/12 + b^2/2, \\ f(4) &= \pi^2/8 + 7a^2/4, & f(5) &= \pi^2/6 + 3c^2/8 + d^2/2. \end{aligned}$$

Therefore

$$-f(2) + f(5) + f(2/5) = d^2 + \frac{1}{2} \mathcal{L}(4/5). \quad (6.17)$$

Inversion and the three-term relation at $3/2$ give

$$f(3/5) = -f(2) + f(3) + f(2/5) - \mathcal{L}(2/3) + \mathcal{L}(3/5).$$

Also $f(4/5) = f(4) - f(5) - \mathcal{L}(1/2) + \mathcal{L}(4/5)$ and $f(3/4) = f(3) - f(4) - \mathcal{L}(1/2) + \mathcal{L}(3/4)$. The Hecke relation at $x = 3/5$, together with (6.8) and (6.17), now becomes

$$J(3/5) = d^2 + T + \frac{1}{2} a(b - c) + \frac{1}{2} a^2 + \frac{2\pi^2}{45}, \quad (6.18)$$

where

$$T = \mathcal{L}(1/2) - \mathcal{L}(2/3) + \mathcal{L}(3/5) - \frac{1}{2} \mathcal{L}(4/5) - \mathcal{L}(3/4) + \mathcal{L}(3/8).$$

The substitution $A = 1/2$, $B = 3/4$ into (6.3) gives

$$\mathcal{L}(1/2) + \mathcal{L}(3/4) = \mathcal{L}(3/8) + \mathcal{L}(1/5) + \mathcal{L}(3/5).$$

Combining this with reflection at $1/3$ and $1/5$ yields

$$T = \mathcal{L}(1/3) - \frac{1}{2} \mathcal{L}(1/5) - \frac{\pi^2}{12}.$$

Finally,

$$\mathcal{L}(1/3) - \frac{1}{2} \mathcal{L}(1/5) = \text{Li}_2(1/3) - \frac{1}{2} \text{Li}_2(1/5) - \frac{1}{2} ab + \frac{1}{2} b^2 + \frac{1}{2} ac - \frac{1}{4} c^2.$$

The cross terms cancel in (6.18); $2/45 - 1/12 = -7/180$ gives the stated constant.

To check the core directly, take its boundary:

$$\partial([1/3] - \frac{1}{2}[1/5]) = \langle 2 \rangle \wedge \langle 3 \rangle - \langle 2 \rangle \wedge \langle 5 \rangle = \langle 2 \rangle \wedge \langle 3/5 \rangle,$$

which agrees with (4.9). The analytic argument above, not this last boundary check alone, supplies the complete constants. $\square$

6.4 All six non-logarithmic formal exceptions are explicit

The known identities for the other two reciprocal pairs are

$$J(1/3) = \frac{\pi^2}{9} + \frac{1}{2} \log^2 2 - \frac{1}{2} \log^2 3 - \text{Li}_2(1/3), \quad (6.19)$$

$$J(1/5) = \frac{7\pi^2}{60} + \frac{1}{2} \log^2 2 - \frac{1}{4} \log^2 5 - \log^2 \phi - \frac{1}{2} \text{Li}_2(1/5). \quad (6.20)$$

These are already present in the manuscript's $J(2/q)$ treatment [1]. For a direct derivation, use (1.2) at $1/3$ and $1/5$, the reciprocal-integer formula

$$F(1/m) - F(1) = -\frac{(m-1)(3m-2)}{24m} \pi^2 - \frac{1}{2} S(m),$$

the three-term consequence

$$F(2/3) - F(1) = -\frac{5\pi^2}{36} + \text{Li}_2(2/3) + \frac{1}{2} \log^2(3/2) - \frac{1}{2} S(3) + \frac{1}{2} S(2),$$

and the displayed $F(2/5)$ formula. The elementary values of S and Euler reflection at $1/3$ and $1/5$ give (6.19)–(6.20) without a branch change.

Equations (6.15), (6.19), (6.20) and reciprocity evaluate every argument in (4.5). Their completeness as a list of formal exceptions is the new classification; the first two classical evaluations are not newly asserted discoveries.

Corollary 6.4 (Cancellation among the three exceptional ratios).

$$J(1/3) - J(1/5) + J(3/5) = \frac{1}{2} \log^2 2 + 2 \log^2 \phi - \frac{2\pi^2}{45}. \quad (6.21)$$

Proof. Substitute the three evaluations. Both rational dilogarithms, the $\log^2 3$ term and the $\log^2 5$ term cancel. The rational π^2 coefficient is $1/9 - 7/60 - 7/180 = -2/45$. $\square$

7 An exact correction to a finite-field assertion

The reviewed arXiv preprint of Radchenko–Zagier, version 2012.15805v1, Section 7.2, printed page 16, includes the finite-field assertion

$$\beta_p(n) = \beta_p(n+1) + \beta_p(n/(n+1)), \quad (7.1)$$

where arguments are reduced modulo the prime p . The same statement appears in the author-hosted preprint checked for this report [2]. For the exterior symbol defined in (2.4), this assertion is false. The issue is not a decimal discrepancy or an unresolved branch convention: the obstruction is a nonzero rationalized exterior symbol.

Proposition 7.1 (Prime-seven counterexample). *At $p = 7$ and $n = 2$, the defect in (7.1) is $3\beta_7(2) \neq 0$.*

Proof. Modulo 7 one has $2/3 = 3$, and $3 \equiv -2^{-1}$. The valid symmetries (2.5) therefore give $\beta_7(3) = -\beta_7(2)$. Equation (7.1) would say $\beta_7(2) = -2\beta_7(2)$.

Here is an independent witness that $\beta_7(2) \neq 0$, not relying on the full kernel theorem. Write

$$a = \langle 1 - \zeta_7 \rangle, \quad b = \langle 1 - \zeta_7^2 \rangle, \quad c = \langle 1 - \zeta_7^3 \rangle.$$

Pairing k and $-k$ in the definition gives

$$\beta_7(2) = 2(b \wedge a + c \wedge b + a \wedge c) = -2(b - a) \wedge (c - a). \quad (7.2)$$

Let

$$A = \log(2 \sin(\pi/7)), \quad B = \log(2 \sin(2\pi/7)), \quad C = \log(2 \sin(3\pi/7)).$$

Since $0 < \pi/7 < 2\pi/7 < 3\pi/7 < \pi/2$, one has $A < B < C$. Take logarithmic absolute values at the two embeddings indexed by 1 and 2. The vectors corresponding to $b - a$ and $c - a$ are respectively

$$(B - A, C - B), \quad (C - A, A - B).$$

Their determinant is

$$-(B - A)^2 - (C - B)(C - A) < 0. \quad (7.3)$$

Thus their exterior product has a nonzero image, proving (7.2) nonzero. The defect is consequently $3\beta_7(2)$, as claimed. $\square$

7.1 Precise correction and unaffected statements

For the definition in (2.4), the universal finite-field three-term identity (7.1) should be removed. Any intended replacement involving an altered module, action or quotient would need a new statement and proof. It is not justified by applying rational Herglotz three-term relations directly to finite-field residues.

The sign and inverse symmetries remain true. The rational boundary formula (2.7) also remains true: it follows directly from finite products, as proved in Proposition 2.2. The current ProveIt manuscript already includes such a direct derivation and explicitly separates it from a finite-field cocycle claim. The new classification therefore does not inherit the erroneous premise.

This correction is deliberately version-specific. It identifies the statement in the preprint pages actually inspected; it does not claim that all editions of the journal article, all subsequent corrections, or every cocycle construction in the paper have been checked. The analytic Hecke relation and the finite dilogarithm evaluations used in Section 6 do not use (7.1) in their displayed proofs.

7.2 Editorial changes recommended for the manuscript

The next integration can replace the open general- $J(p/q)$ formal classification question with Theorem 4.2, while retaining the separate numerical period question. The phrase “modulo $2e$ ” should be preserved in every summary of the mixed-parity criterion; the pair $(8, 3)$ is a useful regression test.

The analytic identity (6.11) and the evaluation (6.15) can be placed immediately after the existing $J(2/q)$ section. The finite-field caveat belongs beside the direct boundary calculation. The theorem ledger should mark the existing kernel theorem, the K -theory facts and the Hecke relation as prior inputs, not new results. No earlier Gaussian or S_p conjecture should have its status changed as a side effect of this integration.

8 Reproducibility, exact tests and numerical diagnostics

The package separates three forms of evidence: the all-parameter ordinary proofs in this article, finite exact regressions for the implementation, and independent high-precision numerical comparisons. None is described as proof-assistant verification. The numeric script is not an interval arithmetic implementation and does not claim rigorous error enclosures.

8.1 Exact finite checks

The script `code/verify_exact.py` uses integer dictionaries for group-ring and exterior-algebra calculations, integer modular arithmetic for classification, and SymPy only for exact polynomial and rational simplifications. The frozen run uses a bound of 500 and records:

Check	Cases
First-difference criterion over odd conductors	50 700
Second-difference criterion over odd conductors	50 700
New dyadic layer four-term group-ring witness	50 828
Odd conductor doubling in a free exterior algebra	2 106
Coprime unordered pairs checked against the recurrence generator	76 115
Height-count regressions at every bound from 1 to 500	500
Leading recurrence-polynomial coefficient checks	62

The first two scans use odd conductors at most 500. The new-layer scan uses even $m \leq 500$ and top conductor $2m$. The free exterior checks use odd conductors at most 101, using the odd lift of a unit residue in the conductor-doubling formula. They verify the expanded identity before any further multiplicative unit relations are imposed. The polynomial tests run through degree 32 in each sign.

The brute-force arithmetic scan finds 781 unordered logarithmic pairs of height at most 500. It finds exactly the three unordered non-logarithmic rational-dilogarithmic pairs $(1, 3)$, $(1, 5)$ and $(3, 5)$. Independent recurrence generation produces exactly the same logarithmic set. The script also checks the prime-seven defect and the rational substitutions in both analytic five-term proofs. Frozen regressions include the failure at $(8, 3)$ and successful cases $(12, 5)$ and $(4, 15)$.

These are meaningful exact cross-checks of implementations: the recurrence generator does not call the modular classifier, and the four-term group-ring calculation does not call a logarithmic numerical rank estimator. Nevertheless their finite ranges cannot establish the infinite classifications on their own.

8.2 Independent defining-integral comparisons

For reduced $p/q > 0$, the numerical implementation evaluates the defining integral after $t = u^q$:

$$J(p/q) = \int_0^1 \frac{qu^{q-1} \log(1 + u^p)}{1 + u^q} du. \quad (8.1)$$

All exponents are nonnegative integers, making this a smooth integrand at zero. The right sides are evaluated independently using log-sine sums or classical dilogarithms, not by another invocation of the same Herglotz identity.

The frozen script `code/verify_numeric.py` performs 38 comparisons at 90 decimal digits and repeats them at 140 digits. There are 23 neighboring cases ($1 \leq n \leq 20$, then 30, 50 and 100),

the five values $J(1/3)$, $J(1/5)$, $J(3/5)$, $J(4/3)$, $J(4/5)$, the reciprocal value $J(5/3)$, the exceptional cancellation identity, and eight further reciprocal pairs. Every residual passes the diagnostic threshold 10^{-d+15} at working precision d .

For orientation, the independently compared value is

$$J(3/5) = 0.30458660277348562359158402248120510653743219820653 \dots$$

The output also records a negative numerical value for the determinant in (7.3). Its rigorous sign is established by the inequality proof, not by this numerical check.

8.3 Files, environment and replay

The recorded environment is Python 3.13.5, SymPy 1.14.0 and mpmath 1.3.0. The Python dependencies are pinned in `requirements.txt`. The principal commands, from the package root, are

```
python -m pip install -r requirements.txt
python code/verify_exact.py --bound 500
python code/verify_numeric.py --dps 90 140
python code/classify.py 3 5
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

An existing TeX installation with the packages listed in `article.tex` is needed for the last command. The scripts do not contact the network after their dependencies have been installed.

The JSON files `data/exact_results.json` and `data/numeric_results.json` preserve the actual runs. `data/logarithmic_pairs.csv` records each pair with its recurrence witnesses, while `data/counts.csv` contains exact larger-height counts. The theorem ledger distinguishes proved results, known inputs and open questions. The integration directory contains an additive manuscript fragment and a proposed bibliography entry; it does not overwrite the existing manuscript or assert that a repository commit has been made.

9 Further research questions

The formal classification closes one stated problem, but it also makes the next tasks more precise. The questions below are not used as assumptions in any theorem of this article.

9.1 Short complete certificates for every recurrence pair

Research question 9.1. Given adjacent terms of an even-coefficient recurrence (5.1) or (5.2), construct the full logarithmic correction for $J(a/b)$ with a replayable branch certificate. Can the number of Hecke and five-term law instances be bounded by a constant times $\log \max(a, b)$ after consecutive pairs are handled by Theorem 6.1?

The classification proves existence by rationalized Bloch descent. It does not automatically make that descent an efficient symbolic algorithm. A useful implementation should keep the finite log-sine sums or cyclotomic products in a compressed representation, and state separately the number of functional-equation steps, the expanded output size and the bit complexity of algebraic-number operations. An $O(\log B)$ descent path does not by itself imply an $O(\log B)$ expanded expression.

The neighboring theorem gives a complete base family. The next test families are the coefficient-2 plus sequence $1, 2, 5, 12, 29, \dots$ and the coefficient-4 minus sequence $1, 4, 15, 56, \dots$. Their exact classification means a failed numerical search should now be treated as a defect of the proposed basis or the certificate construction, not as evidence of formal non-reducibility.

9.2 Other prime filters

Research question 9.2. For a prime $r > 2$, classify the formal boundary of $F(rx) - 2F(x) + F(x/r)$ at all positive rational x , including the cases where the r -adic valuations of numerator and denominator change the cyclotomic fields.

At $r = 2$, the new layer has an order-two kernel and the four-term witness in (3.3) is decisive. For odd r , the analogous new character space is larger and its sparse group-algebra annihilator must be analyzed afresh. Simply replacing 2 by r in (4.3) has not been justified. A first objective is an exact conductor- r distribution identity and a precise old-field intersection theorem.

9.3 Integral structure and torsion

Research question 9.3. What integral lattices underlie the symbol spaces and their dyadic old/new decompositions? Which denominators and torsion classes are necessary in explicit five-term certificates?

Rationalization is essential to the clean criterion (2.2). It deliberately suppresses the torsion structure and branch constants that an integral or flattened Bloch-group implementation must retain. Smith normal forms in a finite presentation could help locate certificate denominators, but a finite matrix must not be mistaken for the entire integral pre-Bloch group.

9.4 Sharper exact counting and uniform depth

Research question 9.4. Can the fixed- R expansion (5.5) be made effective uniformly for R growing with B , and can the remaining floor-function oscillations be isolated by a short exact formula?

The proof gives each fixed fractional-power term and a deliberately coarse tail. A refined treatment should separate the nearly degenerate coefficient-2 minus family from the geometrically growing families, and quantify the inversion of the recurrence polynomials uniformly in their degree. This is a counting problem about the proved formal set, not a numerical-independence problem.

9.5 Period evaluation versus formal obstruction

Research question 9.5. For which explicitly defined subspaces generated by rational Herglotz values is numerical period evaluation injective modulo products of algebraic logarithms and π^2 ? Is there an arithmetic statement strong enough to turn selected nonzero symbols into genuine numerical non-reduction results?

This question is not answered by the present theorem. The six-extra classification is a theorem about the rationalized five-term calculus. Promoting it to an unconditional assertion about all real-number identities would require an additional theorem of a different kind. Any conjecture in that direction should first specify the coefficient field, the permitted logarithmic products and the precise comparison algebra.

9.6 A corrected cocycle formulation and colored extensions

Research question 9.6. Which module and group action, if any, give a valid finite-field cocycle related to the symbols $\beta_p(a)$ while accommodating the explicit prime-seven defect? How does that formulation interact with the Herglotz–Zagier–Novikov parameters of [3]?

A proposed identity must preserve the nonzero wedge detected in (7.3), or explicitly state a quotient that kills it. The rational boundary formula is available independently, so such a reformulation need not be forced into the proof of the classification. Colored parameters may require replacing the real positive Gram matrix by a character-sensitive nondegeneracy argument. This is a distinct extension, not a consequence of the dyadic proof by notation change.

9.7 Formal verification targets

A realistic first formalization would cover the finite group algebra, normalized norm identities, parity-sensitive boundary expansion and Vieta descent. The analytic component could then formalize the Rogers identity on $(0, 1)$ and replay the two substitutions in Section 6. The classical K -theory descent and Dirichlet nonvanishing inputs should remain explicit interfaces until they are themselves available in the chosen proof assistant.

Conclusion

The all-rational formal J problem is governed by a dyadic old/new separation, not by a search over increasingly large baskets of constants. That separation yields a complete modular test, a complete quadratic recurrence parametrization and a hierarchy of exact height asymptotics. The analytic Hecke relation then produces a full infinite identity and a compact exceptional dilogarithm evaluation. The finite-field audit illustrates why direct symbol calculations, branch control and the distinction between formal and numerical statements must remain visible throughout the continuation.

References

- [1] ProveIt contributors, *Polylogarithms and their Arithmetic Bridges*, consolidated manuscript, source snapshot `cc34f73596336f2466d9754cb0f3635bd2bedade`. See the chapter files `09-herglotz.tex` and `10-discovery.tex`, and the report `herglotz-cyclotomic-obstructions`. <https://github.com/VladimirReshetnikov/ProveIt/tree/cc34f73596336f2466d9754cb0f3635bd2bedade/Analysis/Polylogarithms/docs/manuscript>.
- [2] D. Radchenko and D. Zagier, *Arithmetic properties of the Herglotz function*, J. reine angew. Math. **797** (2023), 229–253. The formula numbering and the correction in this article refer to the reviewed preprint `arXiv:2012.15805v1` (2020), especially (15), (22a), (23), (24), Theorem 3 and Section 7.2. <https://arxiv.org/abs/2012.15805v1>. Author-hosted preprint: <https://people.mpim-bonn.mpg.de/zagier/files/preprints/HerglotzFunction.pdf>.
- [3] Y. Choie and R. Kumar, *Arithmetic properties of the Herglotz–Zagier–Novikov function*, `arXiv:2309.10634v1` (2023). <https://arxiv.org/abs/2309.10634v1>.
- [4] D. Zagier, *The dilogarithm function*, in P. Cartier et al. (eds.), *Frontiers in Number Theory, Physics, and Geometry II*, Springer, 2007, pp. 3–65. https://doi.org/10.1007/978-3-540-30308-4_1. Author’s full text: https://people.mpim-bonn.mpg.de/zagier/files/doi/10.1007/978-3-540-30308-4_1/fulltext.pdf.
- [5] F. Calegari, S. Garoufalidis and D. Zagier, *Bloch groups, algebraic K-theory, units, and Nahm’s conjecture*, Ann. Sci. Éc. Norm. Supér. (4) **56** (2023), 383–426. <https://doi.org/10.24033/ansens.2537>. Author’s full text: <https://math.uchicago.edu/~fcale/papers/CGZ.pdf>.
- [6] NIST Digital Library of Mathematical Functions, *Section 25.15: Dirichlet L-functions*, especially Equation 25.15.9, $L(1, \chi) \neq 0$ for nonprincipal χ . <https://dlmf.nist.gov/25.15>.